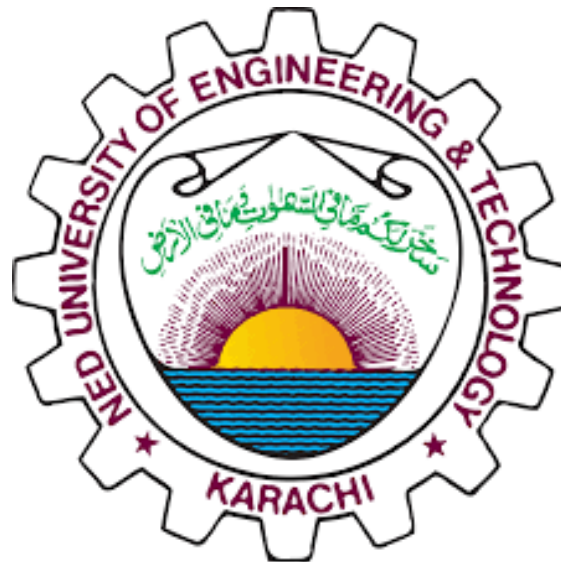




Electric Power Transmission (EE-352)

CEP REPORT

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Abstract

This report is produced to analyze five different geometries of a transmission line using MATLAB GUI. The data collected for each geometry is from either a reference book or published papers. GUI contains the necessary parameters of line be it mechanical or electrical. Generally, the study was conducted after differentiating the voltage levels, structures, conductors, inductance, capacitance, corona loss, ground clearance, sag and stress.

Resultantly, one geometry is proven an efficient and most effective with least cost and loss to environment.

Introduction

What is a Transmission Line?

Transmission lines are the essential parameter of Power System. They are the medium through which Electrical Power is supplied from generation end to the distribution end. Transmission line is a carriageway of current and voltage. It is designed to carry electricity over large distances with minimum losses and distortions. Transmission lines can be classified into two basic categories:

- i. Over Head Transmission line.
- ii. Under Ground Transmission Line.

What are the Parameters of a Transmission Line?

The Parameters associated to the Transmission line are divided into two categories Electrical and Mechanical.

Electrical parameters comprise of; inductance, capacitance, resistance, corona loss, admittance and etc.

While mechanical parameters include sag, tension, ground clearance, stress, strains etc.

The Electrical parameters are uniformly distributed along the length of transmission line. Whereas mechanical parameters are related to the strength and rigidity of tower. For good electric design of transmission line a sound knowledge of these parameters is paramount.

Factors Considered While Designing a Transmission Line:

While designing a transmission line various factors are considered to obtain maximum power transmission. If those factors are neglected then the generated power will not be able to transmit with high accuracy. Following are the factors which are essential while designing a transmission line:

- Cost.
- Reliability
- Power Loss
- Environmental impact
- Durability
- Security

Transmission line Parameters

Inductance

In the selection of a Transmission line Inductance is a very important parameter. In electromagnetism and electronics, inductance is the tendency of an electrical conductor to oppose a change in the electric current flowing through it. The flow of electric current creates a magnetic field around the conductor. The field strength depends on the magnitude of the current, and follows any changes in current. From Faraday's law of induction, any change in magnetic field through a circuit induces an electromotive force (EMF) (voltage) in the conductors, a process known as electromagnetic induction. This induced voltage created by the changing current has the effect of opposing the change in current. This is stated by Lenz's law, and the voltage is called back EMF. Since inductor opposes the flow of current so we must apply some techniques through which its value must be kept as minimum as possible. Following are the formula used for the calculation of the inductance.

a) For Unbundled Conductor:

$$L = 2 \times 10^{-7} \ln (D_{eq} / D_s) \text{ Henry/meters}$$

Where,

D_{eq} = Geometric mean distance

D_s = Geometric mean radius

b) For Bundled Conductor:

$$L = 2 \times 10^{-7} \ln (D_{eq} / D_s^b) \text{ Henry/meters}$$

Where,

D_{eq} = Geometric mean distance

D_s^b = Geometric mean radius for bundled conductor

Additionally,

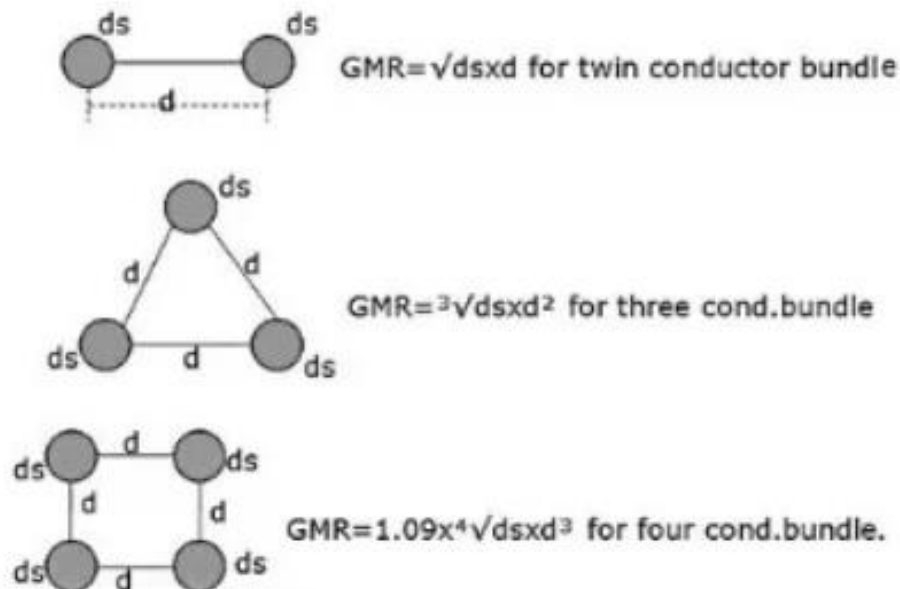
D_s^b for two strand bundle is:

$$D_s^b = \sqrt{D_s \cdot d}$$

And for three strand bundle is:

$$D_s^b = (D_s \cdot d^2)^{1/3}$$

d = distance between bundle conductors



Capacitance

The second most common parameter which is considered is the Capacitance. A capacitor is a device used to store electrical charge and electrical energy. It consists of at least two electrical conductors separated by a distance. (Note that such electrical conductors are sometimes referred to as “electrodes,” but more correctly, they are “capacitor plates.”) The space between capacitors may simply be a vacuum, and, in that case, a capacitor is then known as a “vacuum capacitor.” However, the space is usually filled with an insulating material known as a dielectric. (You will learn more about dielectrics in the sections on dielectrics later in this chapter.) The amount of storage in a capacitor is determined by a property called capacitance, which you will learn more about a bit later in this section. In order to determine the capacitance of a transmission line following formula is used.

$$C = 2 \pi k / \ln (D_{eq} / D_s^b) \text{ Farad/meters}$$

Where,

D_{eq} = Geometric mean distance

D_s^b = Geometric mean radius

Additionally,

For two strand bundle

$$D_{sc}^b = \sqrt{r \cdot d}$$

For three strand bundle

$$D_{sc}^b = (r \cdot d^2)^{1/3}$$

Corona Loss

Corona loss is the other major type of power loss in transmission lines. Essentially, corona loss is caused by the ionization of air molecules near the transmission line conductors. These coronas do not spark across lines, but rather carry current (hence the loss) in the air along the wire. Corona discharge in transmission lines can lead to hissing/cackling noises, a glow, and the smell of ozone (generated from the breakdown and recombination of O₂ molecules). The color and distribution of this glow depends on the phase of the AC signal at any given moment in time. Positive coronas are smooth and blue in color, while negative coronas are red and spotty. Corona loss only occurs when the line to line voltage exceeds the corona threshold. Unlike resistive loss which was due to amount of power lost was a fixed percentage of input, the percentage of power lost due to corona is a function of the signal's voltage. Corona discharge power losses are also highly dependent on the weather and temperature.

$$P = (242.2 / \delta) \times (F + 25) \times (r/d)^{1/2} \times (V - V_c)^2 \times 10^{-5} \quad \text{KW/Km/Phase}$$

Where,

δ = sigma

F = frequency

r = radius

d = spacing

V = supply voltage per phase

V_C = disruptive voltage

Sag

Distance between the highest point of electric poles or towers and the lowest point of a conductor connected between two poles or towers. The difference in level between points of supports and the lowest point on the conductor is called sag. If support level is equal,

$$\text{Sag} = \frac{w \cdot t^2}{8 \cdot T} \quad \text{meters}$$

Where,

W = weight

t = span

T = tension

Stress

Stress is the ratio between the tension present in the transmission line and the cross-sectional area of the cable. Mathematically it can be written as,

$$\text{Stress} = \frac{T}{A} \quad \text{Newton/meter}^2$$

Where,

T = tension

A = cross-sectionals area

Ground Clearance

The minimum distance between bottom conductor and ground is called ground clearance. As per Indian Electricity Rule 1956, Clause No77, the minimum ground clearance. As per this clause, of IE 1956, minimum ground clearance of 33KV non insulated electrical conductor is 5.2 meter. This clearance is increased by 0.3 meter for every 33KV above 33KV.

$$\text{GC} = 5.2 + 0.3048(\text{per } 33\text{kV})$$

Methodology

Analysis of transmission line parameters for five different geometries has been done on five different voltage and structural transmission lines. Two different voltage levels are: 132KV, 220KV and their structure was varied and compared. These geometries can have types like single or double circuit transmission line tower, different selection of conductors and insulators and etc.

The reason behind selecting voltage as our variable parameter is, when the transmission line is designed, the first thing for its structure is operating voltage. Then voltage tells the conductor size and later all the mechanical and electrical parameters are predicted.

Nevertheless, then we have made a GUI on MATLAB by defining every button according to our needs. That GUI has various functions and requires basic data to further calculate the efficiency, regulation and corona loss.

Code of the GUI

```
62 - function CalculateButtonPushed(app, event)
63 -     %code1
64 -     En = app.voltage.Value;
65 -     l = app.length.Value;
66 -     mo = app.mo.Value;
67 -     so = app.so.Value;
68 -     P = app.Power.Value;
69 -     Pf = app.Powerfactor.Value;
70 -
71 -     ht1 = app.ht1.Value;
72 -     ht2 = app.ht2.Value;
73 -     sl = app.Spanlength.Value;
74 -     wt = app.Conductorwt.Value;
75 -     wi = app.Iceloadngwt.Value;
76 -     ww = app.windpressure.Value;
77 -     T = app.Tension.Value;
78 -     %code2
79 -     Dab = app.Dab.Value
80 -     Dbc = app.Dbc.Value
81 -     Dca = app.Dca.Value
82 -     d = app.spacing.Value
83 -
84 -
85 -     switch app.ACSRConductorCodeDropDown.Value
86 -     case 'Drake'
87 -         Ds = 0.0373
88 -         R = 0.0215
89 -         r = 1.108
```

```

90 -         case 'Partridge'
91 -             Ds = 0.0217
92 -             R = 0.0640
93 -             r = 0.642
94 -         case 'Ostrich'
95 -             Ds = 0.0229
96 -             R = 0.569
97 -             r = 0.680
98 -         case 'Hawk'
99 -             Ds = 0.0289
100 -            R = 0.0357
101 -            r = 0.858
102 -         case 'Pheasant'
103 -             Ds = 0.0466
104 -             R = 0.0135
105 -             r = 1.382
106 -         end
107 -         switch app.NoofConductorDropDown.Value
108 -             case 'One'
109 -                 s = 1
110 -             case 'Two'
111 -                 s = 2
112 -             case 'Three'
113 -                 s = 3
114 -             end
115 -         end

116 -         switch app.NoofCircuitsDropDown.Value
117 -             case 'Single'
118 -                 nc = 1
119 -             case 'Double'
120 -                 nc = 2
121 -             end
122 -
123 -
124 -         %Code2
125 -         r = (r/2)*0.0254;
126 -         R = R*0.3048;
127 -         Deq = (Dab*Dbc*Dca)^(1/3)
128 -         d = d^(s-1);
129 -         pow = 1/s;
130 -         Ds = ((Ds*0.3048)*d)^pow
131 -         Dsc = (r*d)^pow;
132 -
133 -
134 -         %code1
135 -         Eo = 21.213*mo*r*so*log(Deq/r);
136 -         En = En/1.732;
137 -
138 -
139 -         Pc = (244/so)*75*(En-Eo)^2*sqrt(r/Deq)*0.00001*1*3;
140 -         app.Pc.Value = Pc;

```

```

142 - R = R*1;
143 - L = (2*0.0000001*log(Deq/Ds))*1000*1
144 - C = ((2*pi*8.85e-12)/log(Deq/Dsc))*1000*1
145 - if nc == 2
146 -     P = (P/nc) + (0.25*P)
147 - end
148 - Ir = (P*1000000)/(3*En*1000*Pf);
149 - c = acos(Pf);
150 - Ir = Ir*exp(-i*c);
151 - theta = angle(Ir);
152 - %
153 - Z = R + 2*pi*L*60*i
154 - Y = 2*pi*50*C*i
155 - Vr = En;
156 - %
157 - Vs = Vr*1000*(1 + (Z*Y)/2) + Ir*Z;
158 -
159 -
160 - Is = Vr*1000*Y*(1 + (Z*Y)/4) + Ir*(1 + (Z*Y)/2);
161 -
162 - sending_end_pf = cos(angle(Vs) - angle(Is));
163 - sending_end_power = 3*abs(Vs)*abs(Is)*sending_end_pf;
164 - Line_efficiency = ((P*1000000)/sending_end_power)*100;
165 - app.Vs.Value = Line_efficiency ;
166 - Vr_noload = abs(Vs)/(abs(1+(Z*Y)/2));
167 - Regulation = ((Vr_noload-(Vr*1000))/(Vr*1000))*100;
168 - app.VR.Value = Regulation ;|

```

```

170 -         if ht1 == ht2
171 -             w = sqrt((wt+wi)^2 + ww^2);
172 -             Sag = (w*s1^2)/(8*T)
173 -             gc = ht1 - Sag
174 -         end
175 -         if ht1 ~= ht2
176 -             h = abs(ht1-ht2);
177 -             w = sqrt((wt+wi)^2 + ww^2);
178 -
179 -             if ht1 < ht2
180 -                 x1 = (s1/2)-((T*h)/(w*s1));
181 -                 x2 = (s1/2)+((T*h)/(w*s1));
182 -                 s1 = ((w*(x1^2))/(2*T));
183 -                 gc = ht1 - s1
184 -             end
185 -             if ht1 > ht2
186 -                 x1 = (s1/2)+((T*h)/(w*s1));
187 -                 x2 = (s1/2)-((T*h)/(w*s1));
188 -                 s2 = (w*(x2^2))/(2*T);
189 -                 gc = ht2 - s2
190 -             end
191 -             app.GroundClearance.Value = gc
192 -         end

```

GUI Interface

Transmission Line Parameters Calculator

Distance b/w phase a&b (m)	8	No of Conductor	Two
Distance b/w phase b&c (m)	8	ACSR Conductor Code	Pheasant
Distance b/w phase c&a (m)	16	No of circuits	Single
Spacing among conductors in case of bundle (m)		0.45	
Line Voltage (KV)	132	Length of transmission line (km)	1
Power factor	0.9	Power (MW)	100
Surface factor	0.85	Air density factor	0.95
Height of tower 1 (m)	30	Height of tower 2 (m)	30
Span length (m)	300	Tension (kg)	0
Conductor wt (kg)	1.5	Ice loading wt (kg)	0
wind pressure wt (kg)	0		

Corona power loss	0	Line efficiency %	0
Ground Clearance at lowest point (m)	0	Voltage regulation %	0

Analysis

1. 132 KV line single circuit bundled conductor

MATLAB App

Transmission Line Parameters Calculator

Distance b/w phase a&b (m)	6	No of Conductor	Two
Distance b/w phase b&c (m)	6	ACSR Conductor Code	Drake
Distance b/w phase c&a (m)	12	No of circuits	Single
Spacing among conductors in case of bundle (m)		0.45	
Line Voltage (KV)	132	Length of transmission line (km)	100
Power factor	0.85	Power (MW)	100
Surface factor	0.9	Air density factor	0.9
Height of tower 1 (m)	40	Height of tower 2 (m)	40
Span length (m)	200	Tension (kg)	1400
Conductor wt (kg)	1.5	Ice loading wt (kg)	1.1
wind pressure wt (kg)	0.9		

Corona power loss	1.468e+04	Line efficiency %	81.7
Ground Clearance at lowest point (m)	30.1737	Voltage regulation %	14.73

2. 220 KV double circuit unbundled

MATLAB App

Transmission Line Parameters Calculator

Distance b/w phase a&b (m)	8	No of Conductor	One
Distance b/w phase b&c (m)	8	ACSR Conductor Code	Ostrich
Distance b/w phase c&a (m)	16	No of circuits	Double
Spacing among conductors in case of bundle (m)		0.45	
Line Voltage (KV)	220	Length of transmission line (km)	100
Power factor	0.85	Power (MW)	100
Surface factor	0.85	Air density factor	0.95
Height of tower 1 (m)	30	Height of tower 2 (m)	30
Span length (m)	250	Tension (kg)	1800
Conductor wt (kg)	1.8	Ice loading wt (kg)	1.2
wind pressure wt (kg)	1.3		

Corona power loss	2.685e+04	Line efficiency %	91.01
Ground Clearance at lowest point (m)	15.81	Voltage regulation %	8.242

3. 220 KV single bundled conductor

MATLAB App

Transmission Line Parameters Calculator

Distance b/w phase a&b (m)	8	No of Conductor	Two
Distance b/w phase b&c (m)	8	ACSR Conductor Code	Pheasant
Distance b/w phase c&a (m)	16	No of circuits	Single
Spacing among conductors in case of bundle (m)	0.45		
Line Voltage (KV)	220	Length of transmission line (km)	100
Power factor	0.85	Power (MW)	100
Surface factor	0.85	Air density factor	0.95
Height of tower 1 (m)	30	Height of tower 2 (m)	30
Span length (m)	250	Tension (kg)	1800
Conductor wt (kg)	1.8	Ice loading wt (kg)	1.2
wind pressure wt (kg)	1.3		

Corona power loss	3.775e+04	Line efficiency %	93.76
Ground Clearance at lowest point (m)	15.81	Voltage regulation %	5.058

4. 220 KV single circuit bundled conductor at unequal heights

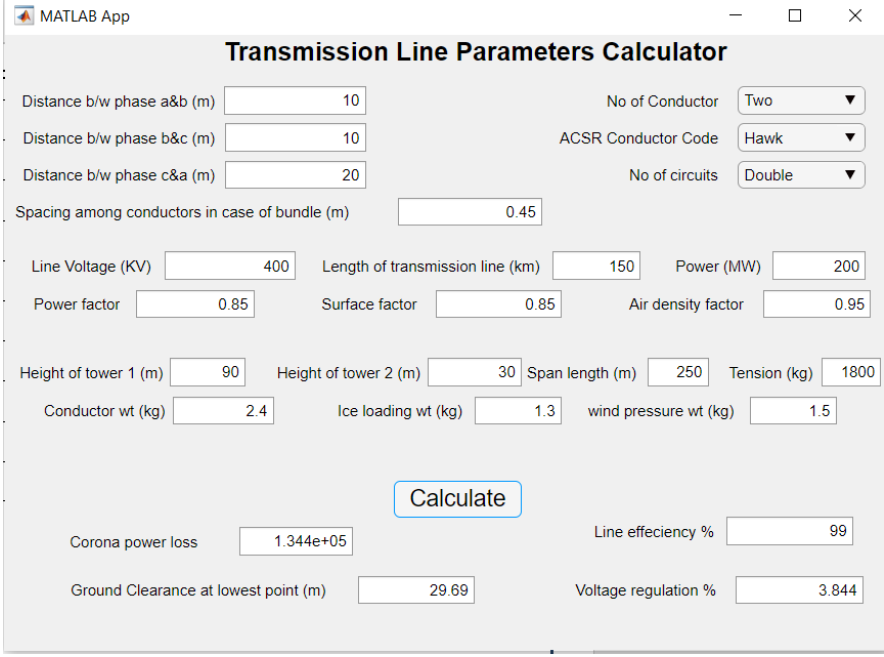
MATLAB App

Transmission Line Parameters Calculator

Distance b/w phase a&b (m)	8	No of Conductor	Two
Distance b/w phase b&c (m)	8	ACSR Conductor Code	Pheasant
Distance b/w phase c&a (m)	16	No of circuits	Single
Spacing among conductors in case of bundle (m)	0.45		
Line Voltage (KV)	220	Length of transmission line (km)	100
Power factor	0.85	Power (MW)	100
Surface factor	0.85	Air density factor	0.95
Height of tower 1 (m)	90	Height of tower 2 (m)	30
Span length (m)	250	Tension (kg)	1800
Conductor wt (kg)	1.8	Ice loading wt (kg)	1.2
wind pressure wt (kg)	1.3		

Corona power loss	3.775e+04	Line efficiency %	93.76
Ground Clearance at lowest point (m)	29.95	Voltage regulation %	5.058

5. 400 KV line double circuit bundled at unequal heights



The image shows a MATLAB App window titled "Transmission Line Parameters Calculator". It contains various input fields and dropdown menus for calculating transmission line parameters. The inputs are as follows:

Parameter	Value
Distance b/w phase a&b (m)	10
Distance b/w phase b&c (m)	10
Distance b/w phase c&a (m)	20
Spacing among conductors in case of bundle (m)	0.45
No of Conductor	Two
ACSR Conductor Code	Hawk
No of circuits	Double
Line Voltage (KV)	400
Length of transmission line (km)	150
Power (MW)	200
Power factor	0.85
Surface factor	0.85
Air density factor	0.95
Height of tower 1 (m)	90
Height of tower 2 (m)	30
Span length (m)	250
Tension (kg)	1800
Conductor wt (kg)	2.4
Ice loading wt (kg)	1.3
wind pressure wt (kg)	1.5

Below the inputs is a "Calculate" button. The results of the calculation are displayed at the bottom:

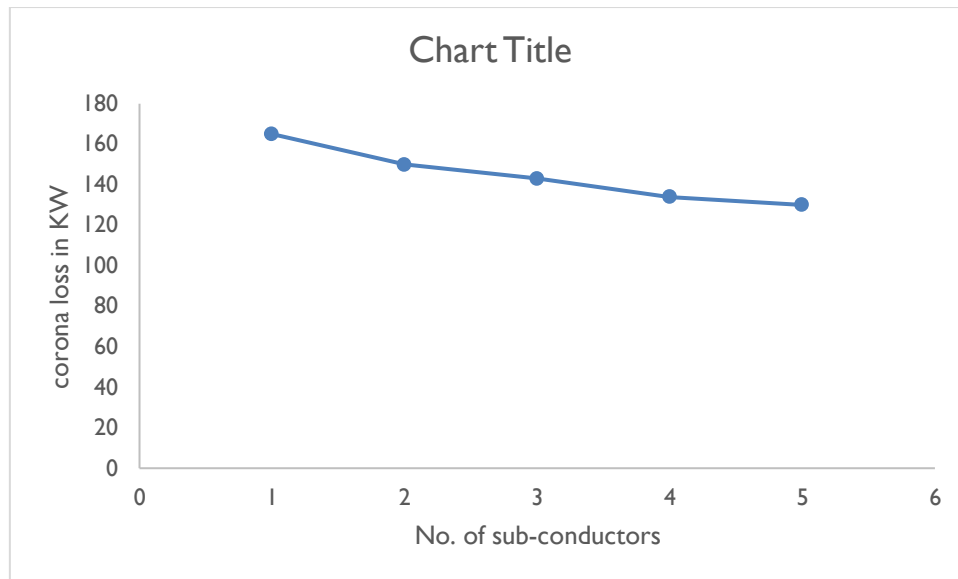
Parameter	Value
Corona power loss	1.344e+05
Line efficiency %	99
Ground Clearance at lowest point (m)	29.69
Voltage regulation %	3.844

Optimization

After reviewing the results of real time data set, we have decided to suggest the amendments for single circuit 132 KV line which is case mentioned above.

Suggestion I: CHANGING THE NUMBER OF SUB-CONDUCTORS PER PHASE

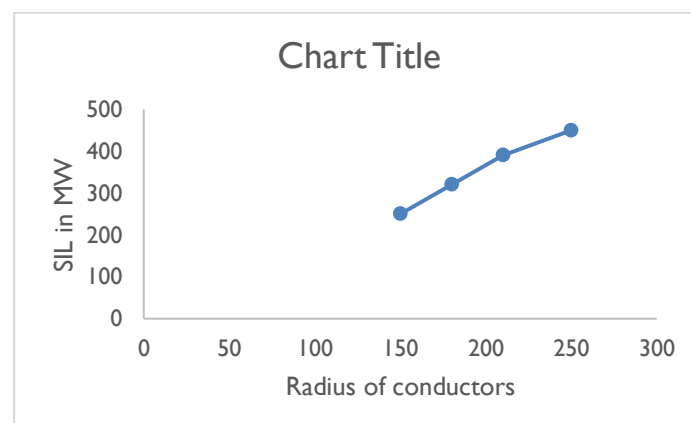
Increase in No. of sub-conductors of bundle increases GM_{Req} of the conductor which would reduce self-inductance of the line and inductance of the line, therefore there will be increment in SIL level is obtained. There is an increment in SIL level observed, so large improvement in power transfer capacity can be achieved. However, increase in No. of sub-conductors per phase loading on existing transmission tower increases, so to reduce the weight shift from twin ACSR conductor to quad ACSR conductor having reduced diameter and weight, but still the overall weight on tower increases, so it is possible to enhance the power capacity of transmission if the tower is designed to carry increase in weight to fulfill the requirement of future increase in power demand.



As the negative gradient of the graph indicates the betterment of power flow with increased number of conductors. We started with 195 KW corona loss in single phase solid conductor and later efficiency is increasing by adding the layers of conductor.

Suggestion 2: SPACING BETWEEN THE BUNDLES

Bundle spacing is the spacing between sub-conductors, as the magnetic field increases Bundle radius R increases and $GMReq$ of bundled conductor increases, which leads to reduction in self-inductance of the line and we can have reduction in line inductance and increase in SIL level as the magnetic field increases. Since, SIL is the MW loading of the line where natural reactive power balance occurs i.e. reactive power produced by a line is equal to reactive power consumed by a line. Therefore, increase in the SIL value would result in the enhancement of efficiency of transmission line and reduced power losses



Conclusion

By applying the above measures we can conclude that the value of inductance is least for the last case i.e. the efficiency/performance of the system maybe increased by increasing number of bundles which reduces the cross-section area of each sub-conductor which in turns reduces skin effect and also it can be improved by increasing the space between conductors which reduces the capacitance hence improves the performance of the transmission line.

Environmental Impacts of Transmission line

Transmission line is built on a large scale affecting many functionalities of this world. Transmission lines have to cross cities, lakes, forest, deserts and etc. which poses an ultimate threat to living species coming in between.

1. Human life sufferings

Humans have to face issues such as distortion in communication signals which interrupts their 2 way communication. The noise of hissing causes pollution and unwanted sounds. Radiating field intensities affects human body. Towers dismantle the scenic beauty and much more.

2. Ecological life sufferings

As the line has to pass forests and green areas, it damages the ecology. The massive tower and its out-coming radiations and field intensities are not suitable for plants.

3. Animal life sufferings

As the forest lands are ruined which are sadly habitat to various animals and after this they are left abandon.

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